

Asymmetric dynamics between supply chain disruptions, oil price shocks, and U.S. investor sentiment[☆]

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ABSTRACT

The COVID-19 pandemic, ongoing Ukraine-Russian conflict, tensions in the Middle East, and US-China trade tensions have severely disrupted global supply chains and the oil market. Amid these distortions, the global supply chain pressures (GSCP) and oil price shocks (OPS) significantly influence investor sentiment. The global shocks are dynamic and asymmetric by nature. Therefore, this research explores the asymmetric impacts of GSCP and OPS on investor sentiment in the United States. We estimate both the short- and long-run associations with monthly data from 1998 to 2023 using a nonlinear autoregressive distributed lag model. After confirming the nonlinearity of the data and preliminary investigations, our long-run estimates reveal that positive GSCP shocks negatively impact investor sentiment, while negative GSCP shocks have insignificant effects. In contrast, positive and negative OPS shocks enhance investor sentiment. This study finds that negative shocks (GSCP-) have a more severe and prolonged impact on investor sentiment than positive shocks (GSCP+), highlighting the need for asymmetry in policy analysis. Similarly, OPS- exerts a stronger influence on investor sentiment than OPS+, with growing disparities over time, emphasizing the asymmetric nature of market reactions to shocks. The findings recommend that policymakers encourage supply chain resilience and diversification strategies to mitigate oil price uncertainty and supply chain disruptions.

1. Introduction

The world economy has recently faced numerous pressures in global supply chains, including congested ports, delayed production operations, prolonged delivery times, higher shipping costs, and border closures. These supply chain strains are provoked amid the COVID-19 containment measures and the Ukraine-Russia conflict (Qin et al., 2023). The economic revival exceeded global commodity market demand and enhanced the GSCP in 2021. Fig. 1 portrays the unstable global supply chain pressures (GSCP) trends, showing spikes in the post-pandemic period and highlighting prompt policy measures to control them. Notably, GSCP significantly influences investor sentiments through multiple channels, lowering confidence and investments (Su et al., 2023). Investor sentiment is an emerging concept that has garnered the attention of policy analysts to unveil its trends and determinants, particularly in the United States (U.S.).

With an extreme GSCP degree, firms' production costs increase due to an adequate supply of raw materials and high shipping charges. It reduces the firm's efficiency gains and competency to meet the rising

market demand. Also, GSCP introduces an uncertain business climate and market volatilities; however, the investors are generally inclined towards a more stable environment. Therefore, investors adopt risk aversion behavior and hedging assets to mitigate potential risks during uncertain times (Sephthorn, 2022). By nature, GSCP is highly sensitive to natural disasters, geopolitical conflicts, climate changes, etc., consequently affecting investor sentiments (Cheng et al., 2023). For instance, the ongoing Russian invasion of Ukraine has imposed sanctions on Russian imports of natural gas and crude oil, producing energy insecurity challenges across Europe and other developing nations. These supply chain stresses instigate investors to diversify energy portfolios. A similar situation was observed during Japan's earthquake in 2011, where the global supply of electronic items and semiconductors was damaged. It illuminates the interruptions in the supply chain of high-tech industries and diverts the investor's decisions in technological investments (Ren et al., 2024). In addition, rising extreme weather and trade tensions can disrupt production, cause shortages, and raise costs. As a result, putting pressure on transport infrastructure and highly integrated supply chains affects investor strategies and sentiments (Cevik

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and Gwon, 2024).

Crude oil prices have also caused GSCP worldwide since the OPEC 1970s. Crude oil is a crucial resource input in economic expansion. Further, it is an investment asset, impacting asset returns and financial markets (Hanif et al., 2024). Thus, Crude oil has two functions: it is a vital component of production processes and an investment asset that assists in diversifying portfolios. Due to its strategic relevance, news concerning the crude oil market and pricing is frequently covered in the media. In addition, oil price variations significantly impact asset returns and financial markets (Naeem et al., 2020). On the supply side, firms experience variations in production costs and revenues, notably affecting the energy sector. In particular, energy firms face profitability fluctuations as oil prices fluctuate. However, on the demand side, consumers' disposable income influences transportation, tourism, and retail purchasing patterns. Oil prices also act as economic indicators, impacting investor sentiment and leading to market volatility. Thus, oil price distortions are critical to determining the associated risk environment and the returns on various assets (Balcilar et al., 2019). In current literature, many studies have investigated how investor sentiments substantially influence oil prices through speculation and economic factors (Sahut et al., 2024; Qadan and Nama, 2018). However, fewer studies explored the impacts of oil prices on investor sentiments. Fig. 2 shows the investor sentiment trends in the United States, denoting complex market structures.

Fig. 3 displays the global oil price shocks (OPS), where high oil market uncertainty and turbulence are observed during the COVID-19 outbreak. The recent pandemic shows that GSCP and OPS are intertwined and influence investor sentiments. Due to border closure, oil prices were slashed, and firms were confronted with distortions in supplying crude oil. In this pandemic outbreak, world economies confront both shocks, which collectively affect investors' sentiments and market conditions (Wang et al., 2022). Investor sentiments further amplify the adverse impacts of OPS and GSCP through behavioral mechanisms (risk aversion) during the shocks. Therefore, policymakers must consider intricate connections between these shocks while formulating economic policies. Due to its financial and commodity attributes, OPS delays essential consumption, production, and investment decisions.

In this situation, OPS is transmitted to financial markets and economic systems, inevitably affecting investor decisions (He, 2020).

Therefore, this study aims to unveil how OPS influences investor sentiments through two channels. Firstly, OPS surges price levels and affects macroeconomic performance, influencing confidence and investor sentiments. For instance, increased price levels are linked with the riskier environment that enhances higher interest rates and inflation, ultimately discouraging investor sentiments. The second channel describes that OPS produces prominent effects on the stock market. Stock market performance significantly affects investor sentiment and decision-making. OPS significantly impacts stock markets, subsequently impacting investor sentiment. In the market, OPS has a spillover effect on investor sentiment. Therefore, investigating how OPS impacts investor confidence in financial markets is essential. Analyzing this complex linkage offers new insights into ascertaining the drivers of investor sentiment, which is crucial because sentiment predicts future economic conditions. Exploring OPS on investor sentiment assists investors in managing risk and hedging methods. It provides implications for policymakers to utilize this information to produce more accurate forecasts and reduce future business cycle fluctuations (Salisu et al., 2019).

In the available literature, numerous studies examined how investor sentiments impact energy price volatility, including crude oil and natural gas. Several studies explored the influence of oil price shocks on stock markets (Hanif et al., 2024; Fu et al., 2024); however, fewer studies have garnered attention to inspecting how oil price shocks determine investor sentiments (Li and Ouyang, 2022; He and Zhou, 2018). Further, many studies have determined the effects of GSCP on macroeconomic indicators such as inflation, commodity prices, and growth indicators (Asadollah et al., 2024; Platitas and Ocampo, 2024), while little has been focused on how GSCP affects investor sentiments. Moreover, these studies employed linear methodologies, such as autoregressive distributed lag (ARDL), structural vector autoregressive (VAR) model, etc. Previous studies have mentioned that crude oil prices have spillover effects on investor sentiments; therefore, applying an asymmetric and nonlinear method is essential in empirical analysis. In the current literature, a potential research gap determines the asymmetric interplay between investor sentiments, GSCP, and OPS in the U.S.

Against the above background, our study expands the existing literature manifold. Firstly, this study elucidates the negative and positive (asymmetric) impacts of GSCP and OPS on investor sentiments. The study used a nonlinear ARDL model using high-frequency (monthly)

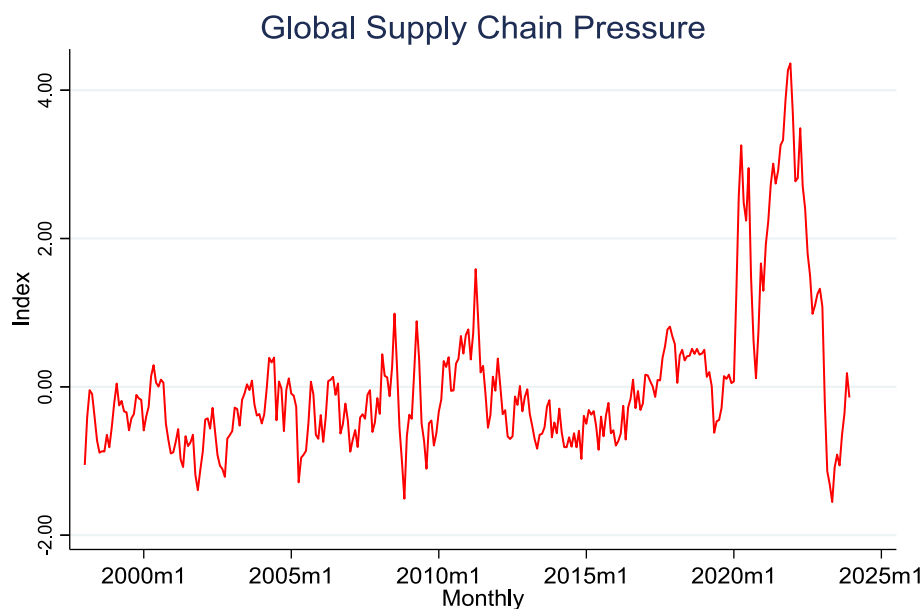


Fig. 1. Global Supply Chain Pressure Index.

Source: Authors' own drawing based on the data from Federal Reserve Bank of New York. <https://www.newyorkfed.org/research/policy/gscpi>

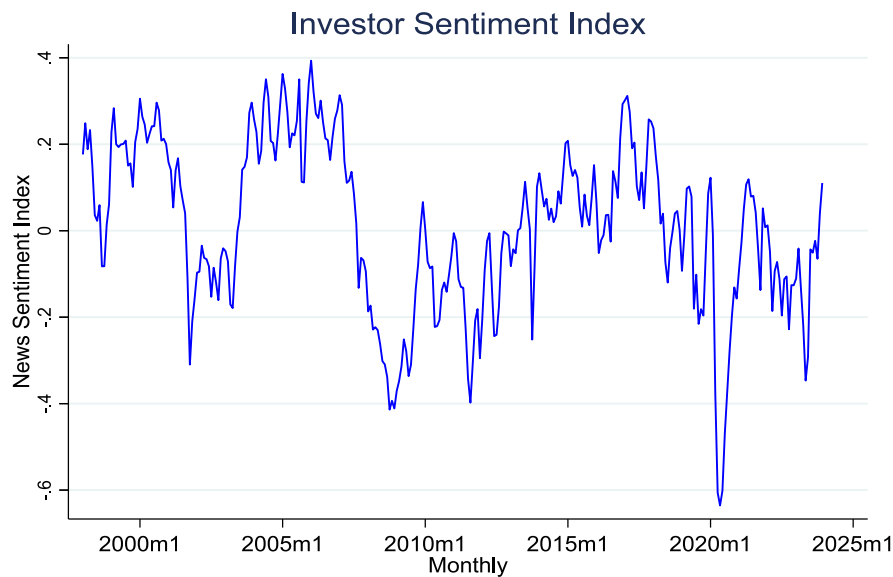


Fig. 2. Investor Sentiment Index.

Source: Authors' own drawing based on the data from Shapiro et al. (2022).

<https://www.frbsf.org/research-and-insights/data-and-indicators/daily-news-sentiment-index/>

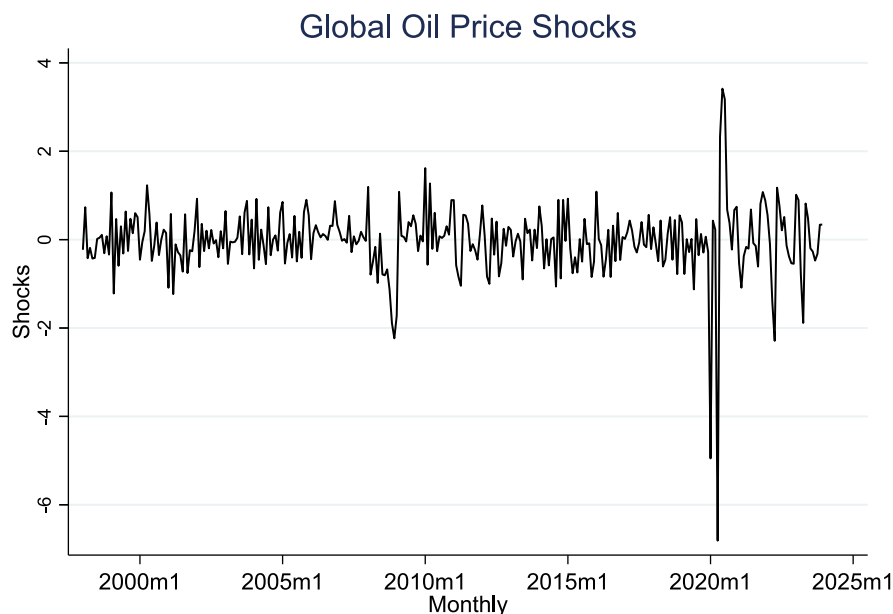


Fig. 3. Trends of Global Oil Price Shocks.

Source: Authors' own drawing based on the data from Baumeister and Hamilton (2019). <https://sites.google.com/site/cjsbaumeister/datasets>

data from 1998 to 2023. Secondly, this research selects the U.S. as a sample space because the U.S. has a dominant position in the international oil market, acting as an oil-importing and exporting nation. As the largest global economy, oil price shocks influence economic growth. In addition, the global supply chain disruptions contributed to U.S. inflation over the last two years. From the beginning of 2021, GSCP has increased input costs, and these supply-side interruptions have accounted for about 60 % of inflation across the U.S. (LaBelle and Santacreu, 2022). The ongoing conflict between Ukraine and Russia drives oil uncertainty. The two Black Swans, namely the COVID-19

outbreak and the Russian invasion of Ukraine, have produced uncertain shocks in the integrated supply chain sector and oil industry. Given the strategic role of the U.S. economy in transferring the shocks to other markets and investor decisions, it is indispensable to focus on the oil market dynamics and supply chain challenges (Hanif et al., 2024). Third, a multidimensional GSCP measure is deployed to capture the supply chain pressures from high transport costs and production fluctuations. This index entails the Harpex index, the Baltic Dry index, and the Purchasing Manager index. The GSCP indicator includes high shipping charges, delayed inventories, backlogs, and air freight costs. It

provides a deep understanding of recent supply chain problems and formulating the confiscated policies. Fourth, investor sentiments are depicted by a daily news-based sentiment index devised by Shapiro et al. (2022). This index is constructed using lexical analysis of news stories from the Washington Post and New York Times. Fifth, the research adopts the nonlinear ARDL model introduced by Shin et al. (2014). Unlike the traditional ARDL technique, the advanced nonlinear ARDL technique allows for decomposing the variable into positive and negative changes (innovations). The nonlinear ARDL model examines whether the positive change in the measure has the same effect as the negative change. In addition, the nonlinear ARDL model's structure facilitates identifying the asymmetric effects in both the long- and short-term.

The remaining study is structured as follows: Section 2 reviews the literature related to investor sentiments and its pertinent factors. Section 3 exhibits the data description and methodology. Section 4 consists of empirical outcomes and discussion, and Section 5 shows the conclusion, policy implications, and the study's limitations.

2. Literature review

A large extent of literature investigated the influence of investor sentiments on predicting oil prices or oil market shocks. For instance, Sahut et al. (2024) identified oil price shocks on Chinese investors' sentiments during the pandemic by adopting the machine learning method. The results show the prominent effects of news-based sentiments during the COVID-19 pandemic and global financial crises (2007–2008). Fu et al. (2024) studied the asymmetric role of investor sentiments on energy prices using the nonlinear ARDL and wavelet approaches between 2000 and 2021. Their analysis exhibits that investor sentiments are inversely related to renewable and non-renewable energy, and according to wavelet results, investor sentiments are significant in the short and medium time frames. Concerning sentiment effects, Rogmann et al. (2024) assessed newspaper and social media effects on consumer energy prices from 2006 to 2021. The estimated outcomes reported that social media has an insignificant impact, while newspaper sentiments significantly increase prices due to high news coverage. Therefore, high emotions in newspaper coverage have marginal price impacts compared to social media. Using high-frequency data, Jawadi et al. (2024) analyzed the nonlinear nexus between volatile energy prices and investor sentiments and documented that both markets show high volatility. Besides, uncertainty and investor sentiments further elevate volatility in commodity prices. Thus, these studies investigated how investor sentiments affect energy prices.

However, scant evidence highlights how energy price shocks, particularly oil price shocks, influence investor sentiments. Li and Ouyang (2022) applied the structural VAR model to analyze the effects of oil price shocks on Chinese investors' sentiments. The outcomes show that oil consumption and supply shocks persist and positively impact investor sentiments; however, aggregate inventory demand shocks temporarily affect them. Applying the advanced dynamic tail quotient correlation method, Lu et al. (2024) analyzed extreme co-movements between three demand shocks and sustainable investments. From tail risk results, upstream and downstream oil price shocks and sustainable investment have time-varying and asymmetric associations and demand shocks pronouncedly affect sustainable investment among all shocks. Hanif et al. (2024) employed a quantile-based approach to evaluate the connection between oil shocks and stock markets in oil-producing and consuming countries. The aim of adopting these techniques is to evaluate conditions in extreme markets under different durations. The authors observed that demand-associated shocks create spillover effects; therefore, tailored investment policies should be framed in expanding investor portfolios and regulating the global oil market. In country-specific studies, Ye et al. (2020) developed an investor sentiment index through principal component analysis and estimated the dynamic nexus between global oil prices and investor sentiments using wavelet

coherence. The findings display that crude oil prices and investor emotions have distinguished movements that change over time and frequency. Le and Luong (2022) discussed a comparative analysis between Vietnam and the U.S. from 2010 to 2020. The scholars applied a time-varying VAR method and found that oil prices and investor sentiments have a dynamic relation and are considered the net transmitters of uncertain shocks in both countries. Akçay (2022) studied the dynamic causality between oil prices and investor sentiments in the U.S. by using the advanced causality approach. The estimates indicate bidirectional causality between them over the different periods.

Crude oil is a critical energy source for economic activities; therefore, fluctuations in oil prices draw a nonlinear relationship with investor sentiments. For instance, He (2020) examined a nonlinear causality between crude oil prices and investor sentiments. In most cases, this relationship is negative and time-varying. He and Zhou (2018) evaluated the nexus between oil price shocks and investor sentiments by adopting time-varying VAR and nonlinear ARDL methods. The study concludes that oil demand side shocks positively influence investor sentiments. Analyzing the Chinese market, Ding et al. (2017) assessed the contagion of global oil prices on investor sentiments. The authors found that any variation in oil prices significantly influences Chinese investor sentiments. Naeem et al. (2023) argued that pandemic-induced oil uncertainty affects investor sentiments. Thus, the emergence of COVID-19 spread oil volatility across the world. Further, the recent Ukraine-Russia conflict has caused spikes in energy prices due to inevitable knock-on effects, increasing energy poverty and household living costs (Zhang et al., 2023). Thus, the above studies confirmed the asymmetric linkage between them by adopting the VAR model, and fewer investigations have adopted the nonlinear ARDL model.

The global supply chain pressure (GSCP) also profoundly impacts investor sentiments after OPS because the supply of crude oil is closely connected with supply chain industries (Alexeev and Chih, 2021). However, the previous studies emphasized the impacts of GSCP on inflation in various countries. For instance, LaBelle and Santacreu (2022) investigated the fact that GSCP caused inflation in the U.S. amid the COVID-19 pandemic crisis. It is imperative to investigate how GSCP influences investor sentiments amid energy shocks. Extant literature emphasizes GSCP with inflation and other growth measures, neglecting its relationship with investor's sentiments. Utilizing Chinese firm-level data, Wang et al. (2022) observed the GSCP effects of COVID-19 containment measures on a firm's financial performance. The author has categorized data into Hubei and non-Hubei firms and found that GSCP impacts Chinese firms while substantial effects are realized for Hubei firms. The severe supply chain restrictions have increased costs, and firms have assumed them to be uncertain global shocks. Meyer (2023) discussed the GSCP's growing impacts on business expectations through the lens of demand and supply side shocks. The analysis reveals that positive demand shocks, such as significant order backlogs, and adverse supply shocks, such as extended supplier delivery times and labor shortages, have increased short-term unit cost expectations and risks. Choudhury et al. (2022) used data from 145 publicly traded enterprises in 21 countries to determine how GSCP causes economic losses measured by shareholder values. The findings show that GSCP deteriorates shareholder value while its negative impacts exceed demand-side fluctuations. Baghersad and Zobel (2021) empirically assessed the extended GSCP influences on United States firms and concluded that different firms reveal varied losses in size and sector after the GSCP occurs.

On the contrary, Seal Jayanta and Paul (2022) used the Indian stock market data and estimated the GSCP and investment behavior after the pandemic lockdowns. The scholars documented that GSCP has no harmful outlook on investors dealing with Chinese suppliers. The ongoing Ukraine-Russia conflict damages the natural gas supply in European countries and produces extreme GSCP problems. In this aspect, Di Bella et al. (2024) researched the economic losses from disruptions in Russian gas supply to European countries. Thus, scant studies focused on

the asymmetric interaction between GSCP and investor sentiments that must be estimated using the advanced nonlinear ARDL approach.

3. Methods and materials

3.1. Data description

This study employs the U.S. monthly data from January 1998 to December 2023. The dataset is high-frequency data, which is available in monthly form. The employment of monthly data efficiently adjusts seasonal variations and addresses point-to-point changes. High-frequency data enhances the reliability and authenticity of model variables used in this study. In our study, the explained variable is the investor sentiment (IS) index measured through the daily news sentiment index developed by Shapiro et al. (2022). The Daily News Sentiment Index is a high-frequency sentiment indicator based on a lexical analysis of news stories about economics from 24 major US newspapers, including the Washington Post and the New York Times. This index reflects the emotions and spirits of investor sentiments in three categories, i.e., positive, negative, and neutral (Jawadi et al., 2024). The data on this index has been sourced from <https://www.frbsf.org/research-and-insights/data-and-indicators/daily-news-sentiment-index/>.

Among explanatory variables, the global supply chain pressure (GSCP) index is used to track the bottlenecks in the supply chain sector. It is a measure that reflects the supply chain restrictions from two dimensions, including higher transport costs and fluctuations in manufacturing production. This composite index consists of the Baltic Dry Index, Harpex Index, and constituents from the Purchasing Manager Index. Benigno et al. (2022) devised this index and retrieved it from the Federal Reserve Bank of New York. Further, oil price shocks (OPS) were constructed by Baumeister and Hamilton (2019), and data were collected from <https://sites.google.com/site/cjsbaumeister/datasets>. The natural gas prices (GP) are measured in U.S. dollars and sourced from the Bloomberg database.¹ Table 1 exhibits the variable description, acronyms, and its sources.

3.2. Theoretical framework and model specification

The interaction between oil price shocks, supply chain disruptions, and investor sentiments is complex and based on various economic and behavioral finance theories. Hamilton (1983) described that oil price

Table 1
Variable Description.

Variables	Acronyms	Description	Sources
Investor Sentiments	IS	Daily news-based sentiment index	Shapiro et al. (2022) https://www.frbsf.org/research-and-insights/data-and-indicators/daily-news-sentiment-index/
Global Supply Chain Pressure	GSCP	Global supply chain index	Federal Reserve Bank of New York https://www.newyorkfed.org/research/policy/gscpi
Oil Price Shocks	OPS	Economic activity oil price shocks	Baumeister and Hamilton (2019) https://sites.google.com/site/cjsbaumeister/datasets
Gas Prices	GP	Natural Gas Prices	Bloomberg database https://www.investing.com/indices/bloomberg-natural-gas-historical-data

Source: Author.

shocks directly influence investor sentiments through macroeconomic indicators, market uncertainty, and corporate profitability. Uncertain geopolitical conflicts (Ukraine-Russia war), supply chain disruptions, and COVID restrictions disturbed economic growth by raising production costs and inflationary pressure. It supports the cost-push inflation theory, in which increasing oil prices reduce investor's profits. Shiller (2003) presented behavioral finance theory and argued that investor sentiments are the psychological response of market participants to these shocks. Adverse oil price shocks increase uncertainty and risk aversion and reduce investments. Herding behavior and information affect the causal relationship between oil price shocks and investor sentiments. Bikhchandani et al. (1992) proposed the information cascade theory, which describes that market participants observe and decide on market trends rather than individual analysis.

In response to heightened media narratives about economic downturns, investors react with a cascade of pessimism in case of negative oil price shocks and supply chain disruptions. Similarly, Kahneman and Tversky (1979) developed the prospect theory, stating that investors adopt loss aversion behavior in response to news about adverse shocks such as supply chain and oil price volatility. Through its effects on trading patterns, speculative activity, and market expectations, investor sentiment has a major impact on oil prices and supply chain shocks. During investor pessimistic sentiments caused by global economic downturns (COVID-19 period), geopolitical threats (Ukraine-Russia conflict), or financial crises, investors lower their exposure to crude oil markets, reducing oil prices. Likewise, supply chain disruptions increase costs and delays, and investor sentiments become negative. Conversely, when economic growth is higher, the oil demand increases, stimulating oil prices. The above behavioral and economic theories explain the causal relationship between oil prices, supply chain shocks, and investor sentiments. Further, these theoretical frameworks describe the reaction of investor sentiments to unpredictable shocks, which amplify the intensity of these shocks. Considering the integrated and dynamic linkage of GSCP and OPS with investor sentiments, the study follows Jawadi et al. (2024) and Li and Ouyang (2022) for model specification. The model's functional form is outlined as follows:

$$IS_t = f (GSCP_t, OPS_t, GP_t)$$
 (1)

The empirical representation can be written as follows:

$$IS_t = \alpha_t + \beta_1 GSCP_t + \beta_2 OPS_t + \beta_3 GP_t + \varepsilon_t$$
 (2)

In the above equation, the intercept term is denoted as α_t , and the parameters of GSCP, OPS, and GP are represented by β_1 , β_2 , and β_3 , respectively. The acronym ε shows the stochastic error, while the letter t indicates the period.

3.3. Empirical strategy

In time series analysis, identifying the presence of unit root is the foremost step. Therefore, this study applied the augmented unit root test developed by Dickey and Fuller (1979). Furthermore, we used the structural-based unit root test of Zivot and Andrews (2002). Both unit root methods identify the integrated properties of desired variables before applying the suitable regression technique. After exploring the variables' stationarity, our research employs a nonlinear ARDL model proposed by Shin et al. (2014), an extension of Pesaran et al. (2001). It estimates asymmetric short and long-run associations between the covariates investor sentiments, GSCP, OPS, and GP using the positive and negative partial sum decomposition. Unlike the traditional ARDL model, a nonlinear ARDL method has certain advantages. It effectively estimates the outcomes in small data (Romilly et al., 2001). The stationary condition is not essential for nonlinear ARDL because it produces unbiased and accurate results whether the variables are stationary at level, first difference, or fractionally integrated. However, a limitation of nonlinear ARDL is that it cannot apply to the integrated variables at the second difference (Meo et al., 2018). Due to its benefits, several scholars

¹ <https://www.investing.com/indices/bloomberg-natural-gas-historical-data>

have adopted this methodology in their analyses (Chen et al., 2020; Li et al., 2024).

The asymmetric long-run regression is considered for this study,

$$IS_t = \gamma^+ UV_t^+ + \gamma^- UV_t^- + \varepsilon_t \quad (3)$$

Where γ^+ and γ^- are the parameters of asymmetric long-run and ε_t is the i.i.d. (independent & identically distributed) process with a mean is zero and finite variance, while UV_t is the symbol in the form of decomposition,

$$UV_t = UV_0 + UV_t^+ + UV_t^-$$

By X_0 is the value of initial uncertainty and $UV_t^+ & UV_t^-$ signifies a partial sum of positive & negative fluctuations in oil rents, digitalization, and energy certainty given by,

$$UV_t^+ = \sum_{i=1}^t \Delta UV_i^+ = \sum_{i=1}^t \max(\Delta UV_i^+, 0)$$

$$UV_t^- = \sum_{i=1}^t \Delta UV_i^- = \sum_{i=1}^t \max(\Delta UV_i^-, 0)$$

The NARDL (p, q) model is developed from a model (1) and can be expressed as,

$$IS_t = \sum_{i=1}^p \alpha_i IS_{t-i} + \sum_{i=0}^q (\delta_i^+ UV_{t-i}^+ + \delta_i^- UV_{t-i}^-) + \varepsilon_t \quad (4)$$

Where, α_i notes as an autoregressive parameter, while $\delta_i^+ & \delta_i^-$ are the distributed lags value of asymmetry, ε_t is the i.i.d process with zero mean and constant variance (homoscedastic process). The equation of asymmetric error-correction can be illustrated as,

$$\Delta IS_t = \beta IS_{t-1} + \delta^+ UV_{t-1}^+ + \delta^- UV_{t-1}^- - \sum_{i=1}^{p-1} \theta_i \Delta IS_{t-i} - \sum_{i=0}^{q-1} \theta_i \Delta IS_{t-i} (\rho_i^+ \Delta UV_{t-i}^+ + \rho_i^- \Delta UV_{t-i}^-) + \varepsilon_t \quad (5)$$

Where, $\omega = \sum_{i=1}^p \alpha_i - 1$ and $\theta_i = -\sum_{j=i+1}^p \alpha_j$, for $i = 1, \dots, p-1$,

Also ρ^+ and ρ^- are the parameters of +ve and -ve short adjustments to changes in the independent variable UV_t , the models mentioned above (3), (4) and (5) are associated by;

$$\sigma = -\delta^+ / \omega \text{ and } \sigma^- = -\delta^- / \omega.$$

The NARDL model proposed by Shin et al. (2014) has been implemented in four steps in eq. (5). Initially, the error correction model is calculated using Standards OLS. Subsequently, the t -test of Banerjee et al. (1998) and the F-test of Pesaran et al. (2001) employ a bound testing approach to search for asymmetric long-run association among the extents of variables are applied to check the cointegration relation with the null hypothesis (no cointegration) and alternate hypothesis (cointegration).

(I): $H_0: \sigma = \delta^+ = \delta^- = 0$ against $H_1: \sigma \neq \delta^+ \neq \delta^- \neq 0$

Afterwards, the long-run and short-run asymmetries are captured by the utilization of the Wald Test. The null hypothesis versus alternate hypothesis is tested for long and short-run asymmetry. Mathematically, it can be written as,

(II): $H_0: \sigma^+ = \sigma^-$ against $H_1: \sigma^+ \neq \sigma^-$

summation against (III): $H_0: \sum_{i=0}^{q-1} \rho_i^+ = \sum_{i=0}^{q-1} \rho_i^-$ against $\sum_{i=0}^{q-1} \rho_i^+ \neq \sum_{i=0}^{q-1} \rho_i^-$

At last, the asymmetric cumulative dynamic multiplier impacts of a unit change in UV_i^+ and UV_i^- on IS_i , are estimated as

$$m_i^+ = \sum_{j=0}^l \frac{\partial IS_{t+j}}{\partial UV_t^+}, m_i^- = \sum_{j=0}^l \frac{\partial IS_{t+j}}{\partial UV_t^-}, l = 0, 1, 2, \dots \quad (6)$$

Where,

$$m_i^+ \rightarrow \sigma^+ \text{ and } m_i^- \rightarrow \sigma^- \text{ as } l \rightarrow \infty$$

Long-run asymmetry is inferred by equality between long-run multipliers

$$Luv^+ = \delta^+ / -\omega \text{ and } Luv^- = \delta^- / -\omega$$

The above methodology discussed is essential for analyzing hypotheses (I-III). Meanwhile, the long-run multipliers (Luv^+ & Luv^-) comply with the,

$Luv^+ = \sigma = -\delta^+ / \omega$ and $Luv^- = \sigma^- = -\delta^- / \omega$ denotes the equilibrium long-run economic growth changes resulting in +ve and -ve uncertainty shocks.

After estimating short- and long-run asymmetric coefficients, the study checks the stability and reliability of estimated parameters through diagnostic estimation. It includes serial correlation, heteroscedasticity, model specification, and CUSUM tests. These diagnostic tests ensure the model's stability after regression analysis.

4. Empirical outcomes and discussion

4.1. Descriptive and unit root analysis

Table 2 presents descriptive statistics of model variables. Gas prices (GP) exhibit the highest mean value, while IS shows a lower mean score. The maximum and minimum values are realized for GSCP and IS, respectively. Moreover, all variables have lower standard deviations, indicating that data is reliable and contains lower dispersions. Also, skewness values are negative, demonstrating that data has a left-sided distribution excluding GSCP. Generally, it assumes that variable series are normally distributed. However, the Jarque Bera test confirms that data has an abnormal distribution. The significant test values underscore that data series are abnormally distributed, which confirms that the NARDL model is suitable for regression analysis. After descriptive statistics, the unit root existence is identified using the ADF and ZA unit root tests. Table 3 reports that variables are stationary at the level, excluding GP, which becomes stationary at the first difference. Thus, none of the variables is integrated at I(2). Similarly, the ZA unit root test affirms the variables' stationarity at I(1) with structural breaks except for IS, which is integrated at level. Both ZA and ADF tests produce unbiased outcomes for all variables.

4.2. Nonlinearity and correlation analysis

Table 4 mentions the results of the BDS test, where all variables from dimensions 2 to 6 are nonlinear. The significant z-statistics at a 1 % significance level indicate that data has nonlinear trends, which confirms that NARDL is a suitable approach for estimating asymmetric coefficients. Table 5 shows the correlation between GSCP, OPS, GP, and IS and their distribution. The correlation matrix prominently reveals a positive relationship between OPS, GP, and IS. In contrast, it illuminates the inverse association between GSCP and IS. In addition, this correlation matrix represents multicollinearity concerns among the data depending on the strength of relationships. Thus, all covariates correlate below 0.6, affirming no multicollinearity problem within the data series.

Table 2
Descriptive Statistics.

Variables	IS	GSCP	GP	OPS
Mean	0.011	0.014	1.773	-0.030
Maximum	0.393	4.361	3.168	3.416
Minimum	-0.635	-1.552	0.017	-6.810
Std. Dev.	0.193	1.009	0.803	0.808
Skewness	-0.470	1.991	-0.604	-2.254
Kurtosis	2.983	7.356	2.679	23.611
Jarque-Bera	11.53***	453***	20.33***	5787***
Probability	0.003	0.000	0.000	0.000

Table 3
Unit Root Test Results.

Variables	IS	GSCP	OPS	GP
ADF(Level)	−4.552***	−3.438**	−17.258***	−2.023
ADF(Δ)	−	−	−	−17.439***
ZA (Level)	−4.061	−4.988***	−17.385***	−2.371
T_b	2007 M3	2020 M01	2008 M02	2008 M10
ZA (Δ)	−11.231***	−	−	−17.491***
T_b	2009 M01	−	−	2013 M12

Note: *** represents a 1 % significance level, and T_b shows endogenous break years.

4.3. NARDL estimations

Before estimating asymmetric parameters, the study confirms the nonlinear cointegration as a prerequisite. For this purpose, the NARDL-bound cointegration test is applied. This test expands the conventional ARDL methodology by estimating the nonlinear cointegration nexus between the variables. The findings of NARDL-bound cointegration are shown in Table 6, which significantly accepts the alternative hypothesis of “cointegration” in the model. The F-statistic of 10.492 surpassed the upper bound critical values at a 1 % significance level. Also, the significant t-statistics is 5.869. Both tests affirm a long-run nonlinear cointegration between OPS, GSCP, GP, and IS.

Table 7 demonstrates the long-run asymmetric associations between the variables. As per the outcomes, the positive supply chain shock (GSCP⁺) significantly negatively impacts investor sentiments. These findings adhere to Choudhury et al. (2022), where increasing supply chain restrictions cause economic losses measured by shareholder values in 21 countries. Likewise, Meyer et al. (2023) verified that upsurging GSCP enhances costs and market risks, reducing investor sentiments in the business environment. However, the adverse shocks in the supply chain sector do not significantly affect investor sentiments. The findings imply that increasing disruptions in the supply chain industry restrict transport systems, delay production operations, and lead to higher shipping costs. It reduces investors' confidence and increases market volatility because investors prefer a sustained business climate (Meyer, 2023). Thus, investors adopt risk aversion attitudes amid an unpredictable supply chain sector. However, negative supply chain shock (GSCP[−]) has an insignificant impact on investor sentiment. Our estimates support the findings of Seal Jayanta and Paul (2022), who found that GSCP has not affected Chinese investor sentiments during the COVID-19 containment measures in India. It suggests to policymakers that investor sentiment is critical when framing economic policies. Therefore, diversified and resilient supply chain policy actions are required to eliminate challenges in the supply chain sector.

The NARDL estimates for positive oil price shocks (OPS⁺) reveal a significant positive relation with investor sentiments. Our estimates concurred with Li and Ouyang (2022), in which the scholars explored the structural positive impacts of oil supply shocks on investor sentiments in China. Primarily, these shocks are transmitted by influencing investor's confidence in the international economic environment and domestic economic fundamentals. The coefficient of negative oil price shocks (OPS[−]) is significantly positive, implying that a negative oil price shock increases investor sentiment. These findings are identical to He and Zhou (2018), in which oil-specific shocks positively associated with investor sentiments in the U.S. In the global oil market, the U.S. is one of

Table 4
BDS test for nonlinearity.

Variables	m = 2	m = 3	m = 4	m = 5	m = 6
IS	34.031***	30.696***	27.653***	25.312***	23.451***
GSCP	24.831***	27.732***	31.520***	36.999***	44.796***
GP	66.834***	89.845***	125.527***	184.129***	281.503***
OPS	7.218***	7.480***	7.697***	7.838***	7.614***

Table 5
Correlation Matrix.

Variables	IS	GSCP	OPS	GP
IS	1			
GSCP	−0.1984	1		
OPS	0.0760	0.0559	1	
GP	0.3036	0.1431	−0.0196	1

Source: Author Estimation

Table 6
NARDL Bound Cointegration test.

Critical values	Lower bound	Upper bound	F-statistics
1 %	3.436	4.758	10.492***
5 %	2.641	3.820	
10 %	2.270	3.372	
k = 4			
1 %	−3.442	−4.822	t-statistics
5 %	−2.858	−4.197	−5.869***
10 %	−2.556	−3.863	

Note: *** shows a 1 % significance level.

Table 7
NARDL Long-run Results.

Variables	Coefficients	t-statistics	Standard errors	P-values
GSCP ⁺	−0.085*	−1.99	0.047	0.047
GSCP [−]	−0.044	−1.25	0.035	0.214
OPS ⁺	0.189***	3.70	0.051	0.000
OPS [−]	0.171***	3.43	0.050	0.001
GP	0.320**	2.87	0.111	0.004
Constant	0.020**	2.28	0.008	0.023
ECT	−0.138***	−5.87	0.023	0.000

Note: *, **, and *** denote 10 %, 5 %, and 1 % significance levels.

the exporting nations that exported 10.15 million petroleum barrels per day in 2023 (EIA, 2023). Thus, increasing shock in oil prices enhances the profitability of oil-producing enterprises and leads to optimistic investor sentiments. Also, these estimates exhibit that speculators consider crude oil to be a financial asset. Thus, investors hedge crude oil against the riskier assets, driving positive investor sentiments in the oil market. These speculators gain profits from oil price fluctuations.

Apart from this, the U.S. imported around 8.51 million barrels per day of petroleum products in 2023. Any negative shock in oil prices reduces production costs and enhances the profits and investor sentiments in oil-importing enterprises. Amid the pandemic crises, the global oil market poses both upscale and downscale shocks in oil prices that significantly affect investors' decisions and emotions. The U.S. economy exports and imports crude oil; therefore, investors' moods towards negative and positive oil shocks are more sensitive. It highlights crude oil market turbulence in the U.S. as the largest economic powerhouse. The government should consider investors' sentiments pertinent to energy insecurity issues. Moreover, energy analysts should regulate oil price fluctuations and promote renewable energy to overcome oil shocks.

As a control variable, gas prices (GP) significantly increase investor sentiments by 0.322 %. The findings mention that increases in natural gas prices contribute to investor sentiments. The estimates are justified because the U.S. has abundant natural gas deposits. An increment in

natural gas prices encourages the confidence of investors in energy markets. The estimates support the outcomes of Rubaszek et al. (2021), who realized the dynamism of the U.S. natural gas market and found that market-based shocks fluctuate natural gas prices, ultimately affecting the investor mood. The error correction term (ECT) exhibits a significant negative coefficient of 0.138, indicating the enduring relationship and dynamic stability among the variables. Thus, any disequilibrium in the short run converges the model towards the steady state by 13.8 %.

Table 8 reveals the short-run estimates of the NARDL model. The coefficient of IS_{t-1} is positive and implies that one unit increase in past investor sentiment (as a lagged variable) substantially influences the current sentiments by 0.207 %. The results suggest that past-year sentiments have influential effects on the present investor moods. Estimates show that $GSCP^+$ negatively affects investor sentiments. On the contrary, $GSCP^-$ has insignificant effects. Thus, both positive and negative shocks in the supply chain are aligned with long-run estimates. The coefficient of OPS^+ does not influence investor sentiments, while OPS^- enhances investor sentiments. This implies that the lowered oil price shocks optimize investor behavior in the short run, as witnessed during COVID-19; however, OPS^+ produces no prominent effects in the short run. In addition, GP has a significant enhancing relationship with investor sentiment and supports long-run estimates.

In comparing long-run estimates, OPS^+ produces more positive effects on investor sentiments than OPS^- . This implies that uncertain oil shocks affect the investors' decisions regarding the oil market, which further contributes to market volatility. The recent pandemic outbreak and the Ukraine-Russia conflict have interrupted the global supply chain sector, which greatly affected investor's strategies and moods. The prolonged delivery times, factory closures, and high shipping costs create uncertainty among investors and make them reluctant for future investment decisions. Thus, the confidence of investors is reduced during positive supply chain shocks, while negative supply chain shocks have no significant effects on investor sentiments. Similarly, global 2008 oil spikes and 1970 OPEC oil prices are real-world examples of OPS. Collectively, GSCP and OPS affect investors' sentiments; however, an asymmetric relationship is found between OPS, GSCP, and investor sentiments, justifying the adoption of the NARDL model. The estimated outcomes provide more coherent policies for the U.S. economy. The U.S. policymakers should monitor oil price fluctuations caused by geopolitical risks, supply chain disruptions, and political risks. In addition, authorities should design long-term diversified and risk mitigation-based energy policies that consider the critical factor of investor sentiments. They should implement early warning alarm systems, risk mitigation tools, and supply chain resilience to control investor sentiments during supply chain and energy disruptions.

4.4. Diagnostic analysis

The study checks the stability and validity of the model through diagnostic analysis. Table 9 shows the Lagrange Multiplier (LM) statistic of the Breusch-Godfrey serial correlation test that contains the null hypothesis of "no serial correlation." The insignificant LM statistic confirms that there is no serial correlation in the model. In addition, the

Table 8
NARDL Short-run Results.

Variables	Coefficients	t-statistics	Standard errors	P-values
IS_{t-1}	0.205***	3.73	0.055	0.000
$GSCP^+$	-0.011*	-2.02	0.005	0.044
$GSCP^-$	0.016	0.91	0.018	0.362
OPS^+	0.007	1.27	0.006	0.205
OPS^-	0.023***	4.11	0.005	0.000
GP	0.025**	2.56	0.009	0.011

Note: *, **, and *** denote 10 %, 5 %, and 1 % significance levels.

Table 9
Diagnostic Analysis.

Tests	Chi 2	P-values
Heteroskedasticity (ARCH)	0.117	0.7322
Breusch-Godfrey LM test	0.025	0.8755
Durbin Watson test	2.00	
CUSUM	Stable	

ARCH test shows that parameters are homoscedastic. Moreover, the coefficient of the adjusted R-square shows a value of 0.873, exhibiting the goodness of the model. This observation suggests that the explanatory variables explain an 87.3 % variation in the explained variable. The parameters' stability is checked by the cumulative sum of recursive residuals (CUSUM) test, and Fig. 4 displays the CUSUM graph that is stable within acceptable bounds at 5 % significance over the period.

4.5. Dynamic multiplier effect

In the NARDL model, this research employed dynamic multipliers to explore the long-run and short-run asymmetrical shocks of GSCP and OPS on investor sentiments. Thus, the multiplier effects exhibit adjustment investor behavior pre and post-effects of OPS and GSCP shocks in Fig. 5. On the left side, the green dotted lines show positive effects, while the red dotted lines indicate the negative impacts of GSCP and OPS on investor sentiments. The blue line represents the asymmetric effects with critical values. Fig. 5 visualizes the adjustment trends for GSCP and investor sentiment nexus using the dynamic multiplier. Positive shocks $GSCP^+$ have a negative impact initially but eventually recover and show trending upward, while negative shocks $GSCP^-$ result in more persistent and pronounced negative effects. The blue asymmetry plot highlights the difference between the two responses. It shows significant divergence, particularly in the short term. Fig. 5 implies that $GSCP^-$ has a more severe and longer-term impact than $GSCP^+$, focusing on considering asymmetry in policy analysis. The right side depicts the asymmetric effects of OPS^+ and OPS^- shocks over time. The upward trend of the dotted green line indicates an increasing positive impact from OPS^+ , whereas the downward trend of the dashed red line indicates increasing negative effects from OPS^- . In comparing OPS^+ , OPS^- substantially influences investor sentiments. The blue asymmetry plot shows a growing disparity between positive and negative shocks. It underscores that policymakers emphasize the asymmetric nature while assessing the repercussions of the shocks.

5. Conclusion and policy implications

5.1. Conclusion

Over the last three years, the world has experienced two major shocks: supply chain restrictions and oil price volatility. These global shocks exert immense effects on investor sentiments. The unpredictable impacts of oil prices and supply chains are asymmetric. Considering the factors, our study provides conducive empirical evidence of asymmetric linkage between oil price shocks (OPS), global supply chain pressure (GSCP), and investor sentiments in the U.S. For empirical analysis, a NARDL method has been applied to estimate positive or negative (asymmetric) short and long-run relationships using the high-frequency data from 1998 to 2023. After the prerequisite confirmation of data nonlinearity, stationary properties, and cointegration among the variables, the study employed the NARDL approach. The long-run findings demonstrate that positive changes in the global supply chain demotivate investors' confidence; therefore, they adopt the risk aversion attitude amid the post-COVID period. In contrast, negative shocks in the supply chain have an insignificant influence on investor's decisions. Besides, the positive and negative shocks in oil prices promote investor's sentiments, highlighting the oil price vulnerability in the U.S. market. In

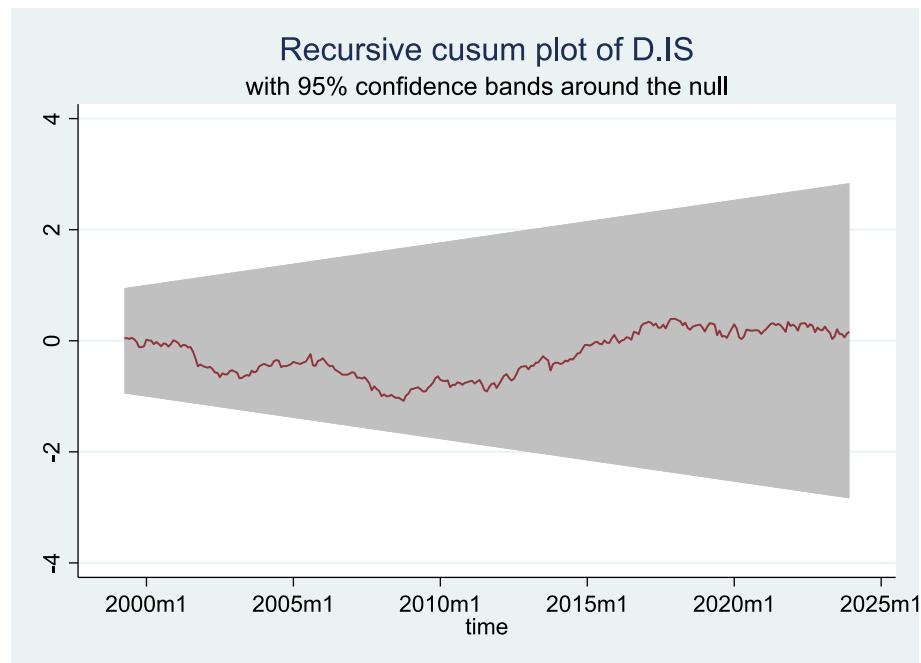


Fig. 4. Plot of CUSUM test.
Source: Author's own generation

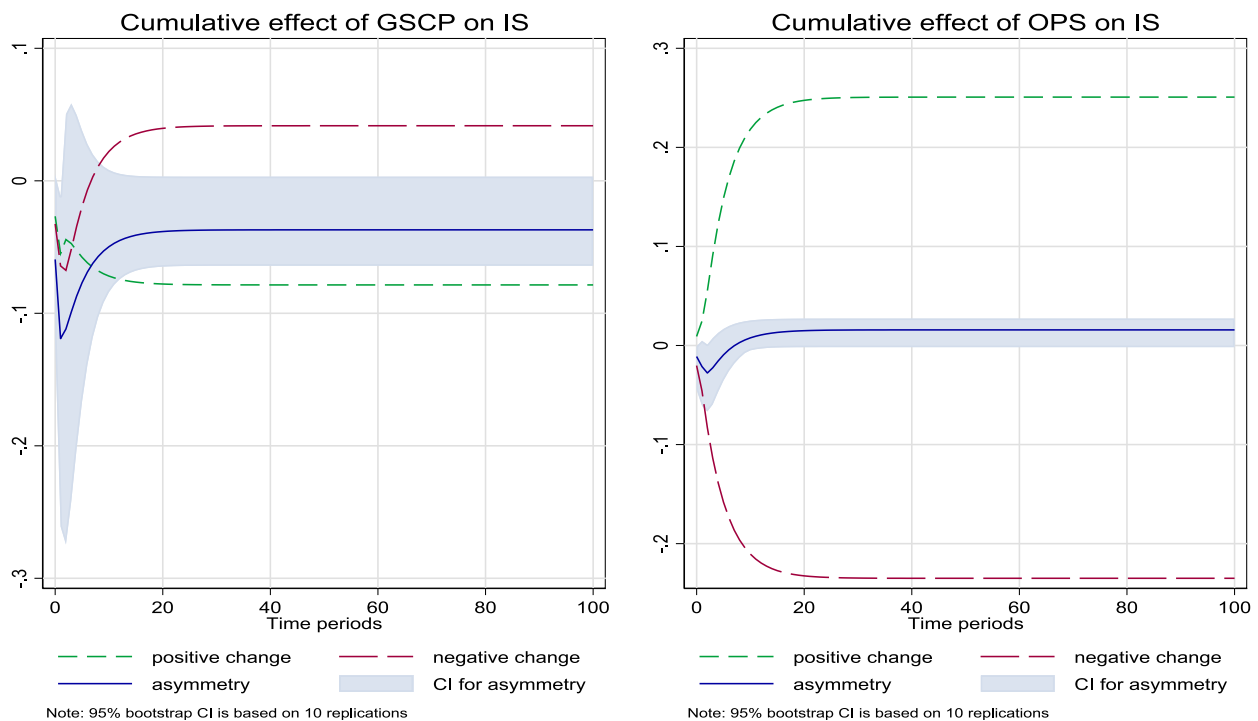


Fig. 5. Dynamic Multiplier Plots for GSCP and OPS.
Source: Author's own generation

addition, the short run confirmed the GSCP long-run estimates. The short-run coefficient of positive OPS aligns with the long-run parameter, while negative OPS is insignificant in the short run. The stability and robustness of the parameters are confirmed by diagnostic analysis.

5.2. Policy implications

Investors should tailor investment portfolio strategies to mitigate oil shocks and market volatility. In this regard, investors adopt a more

granular approach during investment decisions. They should monitor and observe the global oil demand and supply trends using cutting-edge technologies such as artificial intelligence and big data analytics. Moreover, the early warning system predicts the uncertainty in energy markets. Building risk mitigation policies is critical to overcoming oil shocks. Investors should develop diversified and risk management strategies during volatile market conditions. Globally, COVID-19 and the Ukraine-Russia war are responsible for uncertain oil dynamics, while the U.S. economy can transmit the oil price and supply chain shocks to

other markets due to its dominant role in the market. Thus, international players should coordinate policies for economic resilience that assess the potential geopolitical impacts and developments on the oil market. Moreover, the government should establish strategic oil reserves and enhance renewable energy sources as an alternative.

Apart from this, policy analysts should provide data-driven forecasts and transparent and clear information during oil price or supply chain shocks. Enduring economic policies and potential pathways reduce the investors' risks and maintain their confidence in investment-related activities. Similarly, supply chain resilience is required to mitigate the adverse effects on investor sentiments. Prompt and effective responses are imperative for supply chain stability and maintaining a favorable business environment. Diversifying the global supply chain helps to reduce distortions that change investor confidence. It should incentivize enterprises to adopt robust supply chain policies. For instance, the adoption of novel innovations, i.e., automation and localization of critical components, can mitigate the evolving vulnerabilities during uncertainties.

Although this study delved into an intricate relationship between GSCP, OPS, and investor sentiments. Other significant factors may intervene in this complex linkage, such as climate policy shifts, geopolitical tensions, and market behavior. Future studies can investigate these multifaceted factors to inspect investor sentiments in the various regional markets.

CRedit authorship contribution statement

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Appendix A. Supplementary data

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